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Compositional Dependence of Static Shear Viscosity of Immiscible PP/PS Blends

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Phase structures of immiscible polypropylene (PP)/polystyrene (PS) blends with different volume proportions, PP90/PS10, PP80/PS20, PP70/PS30, PP60/PS40, PP50/PS50, PP40/PS60, PP30/PS70, PP20/PS80, PP10/PS90, were observed by means of scanning electronic microscopy (SEM). The zero shear viscosities of the blends were determined according to a modified Carreau model by fitting the curves of static shear rate sweeps of blends tested at 190°C in a Stress Tech Fluids Rheometer. The results showed that the compositional dependence of zero shear viscosity of PP/PS deviated greatly from linear or log-linear additivity. When PS was dispersed in a PP continuous phase, the blends showed negative deviation, while for blends with PP dispersed in a PS matrix, positive deviation was generated. When different theoretical equations of Nielsen, Utracki, Taylor, Frankel-Acrivos (FA), Choi-Schowalter (CS), and Han-King (HK) were used to fit the experimental data of zero shear viscosities of blends, none of them was suitable for PP/PS blends. These experimental phenomena may result from the complex phase structures of the blends and their response to shear conditions, which are discussed in detail and compared with the experimental analysis.

Keywords PP, PS, immiscible blends, static rheology, compositional dependence

1 Introduction

In the past decades, many researchers have been interested in the compositional dependence of zero shear stress viscosity, rubbery plateau modulus or other rheological characteristic parameters of miscible polymer blends^[1–4] and immiscible polymer blends.^[5–7] Generally, whether miscible or immiscible, polymer blends can be grouped into four categories defined by their viscosity-concentration dependence in relation to the linear additivity rule or the log-additivity rule, Eq. (1) and Eq. (2).

$$\eta_b = \sum_i (\phi_i \cdot \eta_i) \quad (1)$$

$$\log \eta_b = \sum_i (\phi_i \cdot \log \eta_i) \quad (2)$$

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where, η_b , ϕ_i , and η_i correspond to the viscosity of blend, and volume fraction and viscosity of ingredient “i” in the blends, respectively.

The four types of polymer blends are: (a) additive blends whose η_b follow Eq. (1) or Eq. (2); (b) PDB (blend showing positive deviation from the linear additivity rule or the log-additivity rule); (c) NDB (blend showing negative deviation from the linear additivity rule or the log-additivity rule); (d) PNDB (blend showing both positive and negative deviation from the linear additivity rule or the log-additivity rule).

In Eq. (1) and Eq. (2), η_b and η_i are often referred to as the zero shear viscosities η_{0b} and η_{0i} of the polymer blend and of the components. However, for a number of polymeric systems, it may not be possible to determine the zero shear viscosities, because a constant level of shear or extensional stress must be used. In addition, with the increase of stress, an initial PDB behavior may change into NDB. Also the compositional dependence of viscosity of a certain polymer pair could be affected by many factors, such as molecular weight of components, molecular interactions in miscible blends, interfacial tension and adhesion of immiscible blends, and so on. The rheological behaviors of polymer blends are so complex that it is difficult to determine and describe the real compositional dependence of viscosity for certain polymer blends, especially for immiscible blends.

For different polymer blends, different researchers have provided specific theoretical equations to describe the compositional dependence, which can be divided into two groups: empirical equations and equations related to phase morphology. Most of the provided theoretical equations were designed for binary blends.

The linear additivity rule shown in Eq. (1) can be treated as the simplest empirical equation. The log-additivity rule shown in Eq. (2) has been the most widely used. In 1974, Nielsen^[8] described a model, resulting in Eq. (3), which could describe PDB (when $n > 1$) or NDB (when $n < 1$). When $n = 1$, the Nielsen equation can be regarded as a linear additivity rule. However, the Nielsen equation is not suitable for PNDB.

$$\eta_b^n = \phi_A \cdot \eta_A^n + \phi_B \cdot \eta_B^n \quad (3)$$

Utracki^[5] pointed out that in immiscible polymer blends the viscosity-concentration dependence was determined by two mechanisms: an interlayer slip and the emulsion-like viscosity increase culminating at the inversion concentration. He derived Eq. (4) which provides insight into the flow mechanism, $\log \eta_L$ and $\Delta \log \eta^E$ are given by Eqs. (5) and (6). $\log \eta_L$ is due to the NDB behaviors because of interlayer slip and β is the interlayer slip factor. $\Delta \log \eta^E$ is an added term derived from the concept of the emulsion-like behavior of polymer blends and $\eta_{\max}$ is a parameter determining the magnitude of the PDB effect. Generally, the Utracki equation can be used to describe PDB, NDB, and PNDB behaviors when suitable values of β and $\eta_{\max}$ are selected experimentally.

$$\log \eta_b = \log \eta_L + \Delta \log \eta^E \quad (4)$$

$$\log \eta_L = -\log\{1 + \beta[\phi_A(1 - \phi_A)]^{1/2}\} - \log\left(\frac{\phi_A}{\eta_A} + \frac{1 - \phi_A}{\eta_B}\right) \quad (5)$$

$$\Delta \log \eta^E = \eta_{\max} \left[1 - \frac{(\phi_A - \phi_{AI})^2}{\phi_A(1 - \phi_{AI})^2 + (1 - \phi_A)\phi_{AI}^2} \right] \quad (6)$$

Considering the phase morphology of polymer blends, Taylor^[6] provided Eq. (7), in which η_b is the viscosity of blend, η_m the viscosity of matrix, p the ratio of shear viscosities of the

dispersed phase and the matrix, and ϕ the volume fraction of the dispersed phase. Based on further considerations to reveal the dependence of η_b on the fine phase structures, Choi and Schowalter (CS)^[7] developed Eqs. (7) and (8). But in these equations, the effects of shear conditions were ignored completely.

$$\eta_b = \eta_m \left(1 + \frac{5p+2}{2p+2} \phi \right) \quad (\text{Taylor}) \quad (7)$$

$$\eta_b^{\text{CS}} = \eta_m \left\{ 1 + \frac{5p+2}{2p+2} \phi \cdot \left[1 + \frac{5(5p+2)}{4(p+1)} \phi \right] \right\} \quad \text{CS} \quad (8)$$

With the deformation of dispersed phase in polymer blends or shear conditions being considered, Frankel and Acrivos (FA)^[9] provided an equation on the basis of the Taylor equation, shown in Eq. (9), where $\dot{\gamma}$ is the shear rate related to the deformation of droplets of the dispersed phase, R the radius of the droplets, Γ the interfacial tension between the two components of the blend. And similarly, Han and King (HK)^[10] provided Eq. (10) on the basis of the CS equation.

$$\eta_b^{\text{FA}} = \frac{\eta_m}{1 + \Lambda^2 \text{Ca}^2} \left\{ 1 + \frac{5p+2}{2p+2} \phi + \Lambda^2 \text{Ca}^2 \left[1 + \frac{5p+2}{2p+2} \phi - \frac{19p+16}{2(p+1)(2p+3)} \right] \right\} \quad (\text{FA}),$$

where,

$$\Lambda = \frac{(2p+3)(19p+16)}{40(p+1)}, \quad \text{Ca} = \frac{\eta_m \dot{\gamma} \cdot R}{\Gamma} \quad (9)$$

$$\eta_b^{\text{HK}} = \eta_b^{\text{CS}} \frac{1 + \Lambda_1 \Lambda_2 \dot{\gamma}^2}{1 + \Lambda_1^2 \dot{\gamma}^2} \quad (\text{HK}),$$

where,

$$\begin{aligned} \Lambda_1 &= \text{Ca} \frac{(19p+16)(2p+3)}{40(p+1)} \left(1 + \phi \frac{5(19p+16)}{4(p+1)(2p+3)} \right) \\ \Lambda_2 &= \text{Ca} \frac{(19p+16)(2p+3)}{40(p+1)} \left(1 + \phi \frac{3(19p+16)}{4(p+1)(2p+3)} \right) \end{aligned} \quad (10)$$

Because the Taylor equation and the CS equation used simple functions to describe the complex interphase relations, with responses of phase structures to shear conditions being ignored, the four equations can not reflect the real mechanisms of polymer blends. Though the FA and HK equations introduced shear conditions into the systems, they were similar to the former Taylor and CS equations, and were empirical equations to some degree. Additionally, the FA and HK equations were only suitable for typical immiscible blends with sphere-shaped dispersed phase in a continuous phase. However, they were not suitable for immiscible blends with co-continuous phase structure or miscible polymer blends without specific phase structures, in which R and Ca could not be determined easily.

In this paper, typical immiscible PP/PS blends with different compositions were prepared. The phase morphology of the blends was obtained by scanning electron microscopy (SEM). At 190°C the zero shear viscosities of the blends were calculated using the modified Carreau model by the static shear rate sweep method.^[11,12] We used

different theoretical equations designed for compositional dependence of polymer blends to fit the resultant experimental data. The complex relations between the rheological behaviors of immiscible polymer blends and the related typical phase structures are discussed in detail, with the possible responses of phase structures to shear conditions being focused on.

2 Experimental

2.1 Materials and Blending

Polypropylene (PP) and polystyrene (PS) were of industrial grade, from Beijing Yanshan Petrochemical Corporation. The molecular weights ($\bar{M}_n$) were 3.1×10^5 for PP1300 and 1.2×10^5 for PS666D, respectively. After being thoroughly dried under dynamic vacuum for at least four hours at 60°C, PP and PS with different volume proportions were melt mixed at 210°C in an XSS-30 mixer for 20 min with the rotation speed of 32 rpm; the obtained blends were cooled to room temperature for further investigations.

2.2 Morphology Observed by SEM

In order to obtain the typical PP/PS phase morphology of the samples to relate to the rheological properties, samples used for SEM observations were cut from $100 \times 100 \times 1$ mm sheets molded at 190°C with the pressure of 20 MPa, from regions close to the part prepared for rheological measurements. The samples were fractured after being cooled in liquid nitrogen for 40 min, and then gold was sprayed on the fresh surface. Phase morphology was observed by a Philips XL-30 SEM and typical images were selected carefully to represent the phase structures.

2.3 Rheological Measurements

Rheological measurements were performed with a Stress Tech Fluids Rheometer at 190°C using parallel plate geometry with the gap being set at 0.5 mm, and the shear rate sweep mode selected to obtain $\eta_b \sim \dot{\gamma}$ relations for calculations of zero shear stress viscosity of PP/PS blends by the modified Carreau model. Sample disks with diameter of 20 mm and thickness of 1 mm were cut from a sheet, the dimension of which was $100 \times 100 \times 1$ mm³. The rheological measurement was started after 20 min of thermal balance for each sample to get a stable experimental condition.

3 Results and Discussion

3.1 Morphology of PP/PS Blends

For typical immiscible polymer blends of PP/PS, the rheological behavior is closely connected to the phase structures. Thus the SEM observations were first carried out to determine whether cryogenically fractured droplets or co-continuity existed. Results can be seen in Fig. 1. In order to make clear the details of finer phase structures, micrographs with different magnifications were selected. All micrographs showed a strong incompatibility between the two polymer phases. Sharp phase boundary can be found both in the cases where the polymer blends contain, for example, 10 vol.% of either PS or PP as the minor phase, and in the cases where co-continuity tends to form in blends with volume composition of PP50/PS50. Thus the spheres visible in images of blends with

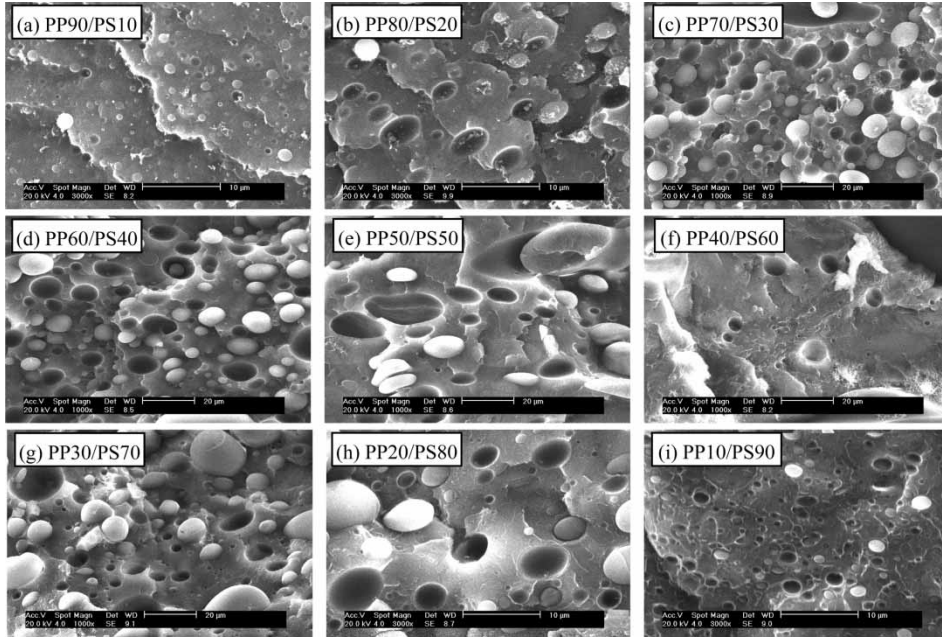


Figure 1. SEM images of PP/PS blends with different compositions.

PS or PP as the minus phase with volume proportion less than 50 vol.% represent PS or PP dispersed component. The results observed were the same as the typical SEM images for PP/PS blends obtained by Funke et al.^[13] The sharp interface may be attributed to the large interfacial tension between PP and PS.

Based on Fig. 2, the numerical average diameters (d_p) for blends with PS or PP dispersed in the other PP or PS matrix phases as droplets were calculated through a software developed in our laboratory. Meanwhile, the specific interfacial surface areas (S_p) of the blends were obtained based on Eq. (11).

$$S_p = \frac{6\phi_d}{d_p} \quad (11)$$

3.2 Determination of η_0

In several theoretical equations, the zero shear viscosity (η_0) is always the most important rheological characteristic parameter selected for polymer blends relative to the compositional dependence. The zero shear viscosity can be either directly obtained by inspection of the experimental data or else estimated from the relaxation spectrum.^[4] The inspected experimental data can be either static shear rate sweep curves or dynamic frequency sweep curves. Here, to determine the zero shear viscosity of PP/PS, the modified Carreau model, shown in Eq. (12), was employed to fit the static shear rate sweep curves, where η is the viscosity, $\dot{\gamma}$ the shear rate, τ^* a constant determined from the intersection of the Newtonian viscosity asymptote and the power-law region asymptote, and n the “power –law index”.

$$\eta = \frac{\eta_0}{(1 + \eta_0 \dot{\gamma} / \tau^*)^{(1-n)}} \quad (12)$$

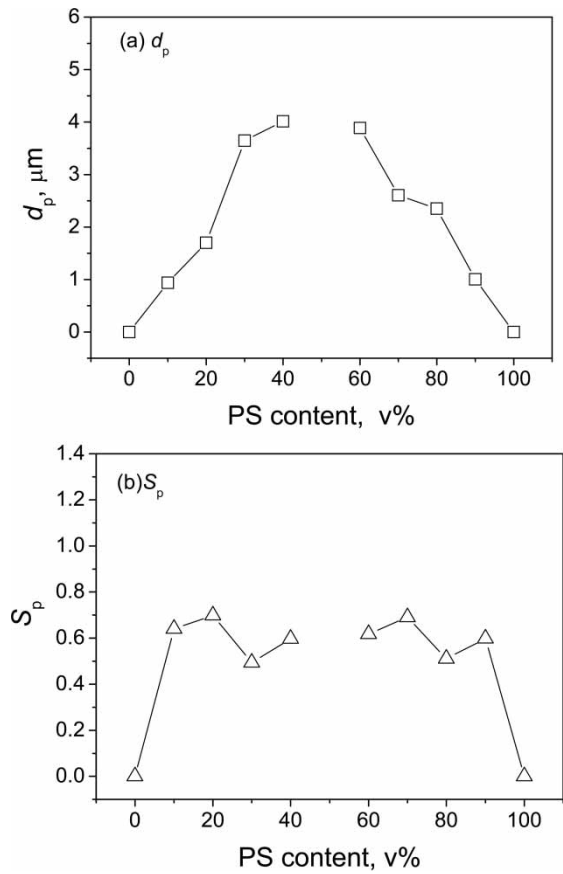


Figure 2. Compositional dependence of d_p and S_p of PP/PS blends (SEM).

Figure 3 shows the shear rate sweep curves of PP/PS with typical compositions. From these curves, zero shear viscosity and some parameters in the modified Carreau model could be determined. Results are shown in Table 1.

As seen in Fig. 4, the zero shear viscosity of PP/PS showed large positive deviation when PS was dispersed in a continuous PP phase, whereas it showed small negative deviation where PP was dispersed in a continuous PS phase. The compositional dependence of zero shear viscosity undergoes a transition according to the phase structure evolution shown in Fig. 1.

Additionally, compared with blends in which <50 vol.% PS was dispersed in a continuous PP phase, blends in which <50 vol.% PP was dispersed in a PS matrix showed different rheological behaviors; though their phase structures were almost the same, the zero shear viscosities of the phases changed significantly. From Fig. 3 it is known that the shear viscosity of PS at 190°C is much larger than that of PP. Thus the droplets of PS dispersed in a PP matrix can be regarded as a hard phase. The droplets of PP dispersed in a PS matrix can be regarded as a soft phase. This means that soft-in-hard phase structure tended to provide larger positive deviation, while hard-in-soft phase structure tended to provide larger negative deviation. The different rheological behaviors must be due to the resultant different response of the related phase structures

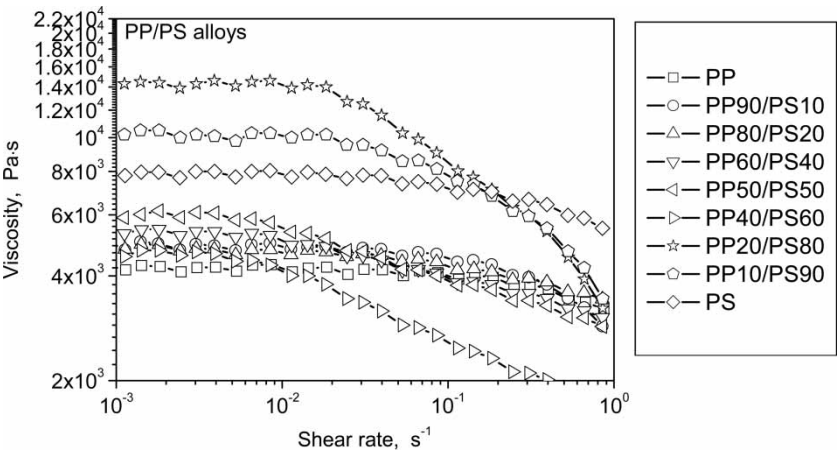


Figure 3. Shear rate sweep curves of PP/PS blends with different compositions.

of blends with different compositions, or even with the same composition but with different soft-in-hard or hard-in-soft types of phase structures.

3.3 Theoretical Analysis of Compositional Dependence of η_0

In order to obtain better understanding of the compositional dependence of the zero shear viscosity of our polymer blends, several different theoretical equations were used to fit experimental data.

As an important experimental theoretical equation, the Nielsen equation can be used to describe positive or negative deviations. Fig. 5 shows the fitted results with n being set as different values for the Nielsen equation. When $n = -3$, the Nielsen equation fitted the experimental data very well only when PS was dispersed in a PP matrix. And when n increased, the Nielsen equation deviated from the experimental curve more obviously.

Table 1
Parameters in a modified Carreau model determined for PP/PS blends

PS content, v%	Zero shear viscosity (η_0), Pa · s	Critical shear stress (τ^*), Pa	Power law index (n)
0	4600	524.4	0.8842
10	4955	438.5	0.8080
20	4920	315.9	0.8992
30	4933	72.4	0.9039
40	5463	62.3	0.8886
50	6159	32.4	0.8545
60	4741	24.8	0.7951
70	15560	175.8	0.5746
80	14390	264.8	0.6429
90	10470	321.4	0.7457
100	7958	314.3	0.8877

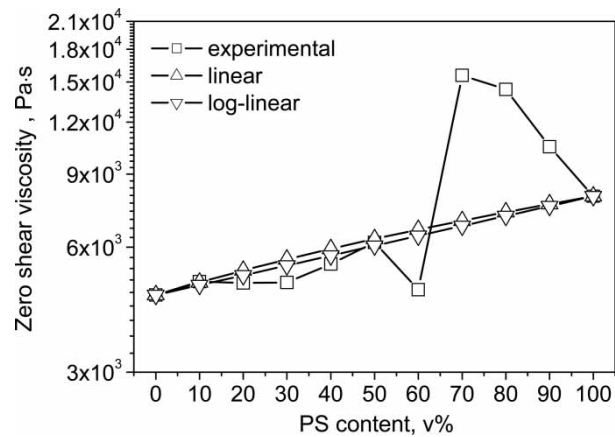


Figure 4. Dependence of zero shear viscosity of PP/PS blends on composition.

For blends in which PP was dispersed in a continuous PS phase, the Nielsen equation showed an obvious difference to the real experimental result. In addition, the Nielsen equation can not be used to describe the compositional dependence of PNDB behavior of PP/PS blends with the parameter n being set as a certain value. Thus it can be concluded that use of the simple Nielsen equation to provide descriptions of rheological behaviors of blends was not suitable. The complexity of rheological behaviors of polymer blends should be related to the phase structure evolution of the blends.

In the Utracki equation, the compositional dependence of zero shear viscosity is closely related to the phase structure evolution of polymer blends with different compositions, with two parameters of β and $\eta_{\max}$ introduced. β and $\eta_{\max}$ represented interlayer slip and the emulsion-like viscosity increase culminating at the inversion concentration, respectively. But the two parameters have no precise physical meaning and can not be easily determined. Fig. 6 shows the fitted curves based on the Utracki equation with values of β and $\eta_{\max}$ being selected empirically. For curves with $\beta = 0$ in Fig. 6a,

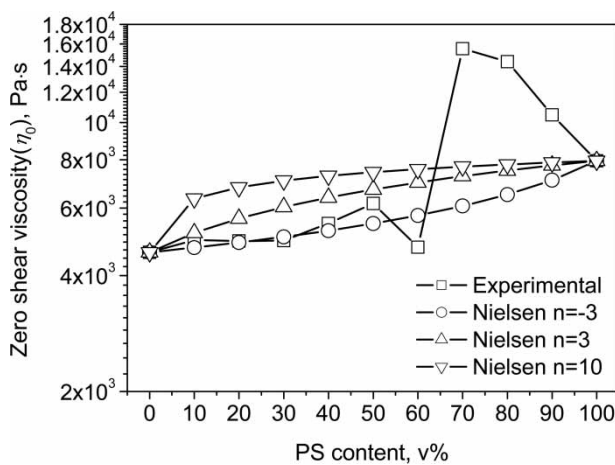


Figure 5. Fitted results with n set as different values for the Nielsen equation.

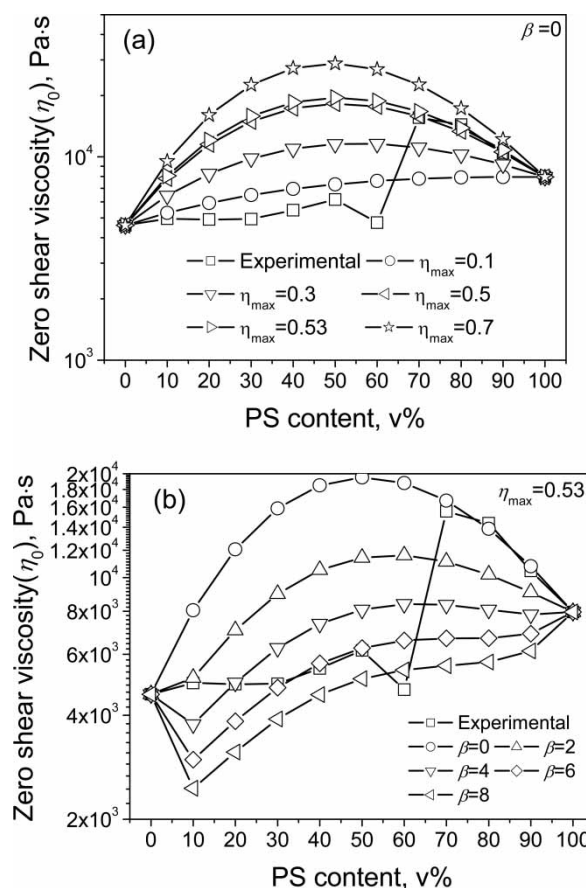


Figure 6. Fitted results from the Utracki equation with β and $\eta_{\max}$ set as (a) $\eta_{\max}$ is alterable and $\beta = 0$, (b) β is alterable and $\eta_{\max} = 0.53$.

there is no interlayer slip in the blends according to the Utracki equation, and all curves showed positive deviations whose amplitude was due to the value of $\eta_{\max}$. Thus, for blends with PS being dispersed in a PP continuous matrix, the Utracki equation with $\beta = 0$ provides larger zero shear viscosity than the experimental data which shows negative deviation characteristics. For blends in which PP was dispersed in a PS matrix, the Utracki equation fitted better. When $\eta_{\max} = 0.53$, the fitted curve covers the experimental curve completely. In Fig. 6b, $\eta_{\max}$ was set as 0.53 and β was changed from 0 to 8, implying the increase in interlayer slips. The results show that the Utracki equation can fit only part of the experimental curve, with given values of parameters of β and $\eta_{\max}$. When β is set small, the Utracki equation fitted well when PP was dispersed in PS and deviated much from the experimental data when PS dispersed in PP. If β was set larger, the Utracki equation may fit well when PS was dispersed in PP, but deviate much from the experimental data when PP was dispersed in PS. So, for PP/PS blends with PS content of 0 to 100 vol.%, the compositional dependence of zero shear viscosity can not be described well by the Utracki equation with a single β and $\eta_{\max}$. This may be due to the absence of phase morphology and ratio of zero shear viscosities of components in the Utracki equation.

The Nielsen and Utracki equations are empirical theoretical equations, though in the Utracki equation, characteristics of the phase structures is concerned to some degree but not enough. In the Taylor and CS equations, the dispersed phase in blends is treated as uniform spheres, on the basis of which different equations are provided to describe the variations of the zero shear viscosity of blends with the ratio of zero shear viscosity of the components (η_0) being introduced into the equations. And in fact, the CS equation can be regarded as a modified Taylor equation, with concern for more subtle details about the phase structure. Both the Taylor and CS equations are not suitable for blends which tend to have co-continuous phase structure. The effect of average diameter (d_p) of the dispersed phase and other factors on the compositional dependence of the zero shear viscosity of blends is included by specific mathematical equations. As shown in Fig. 7a,b, zero shear viscosities predicted by the CS equation are larger than those predicted by the Taylor equation for all blends. When PS was dispersed in a PP phase, both equations showed obvious positive deviations; the Taylor equation was nearer to the experimental data. When PP was dispersed in the PS phase, the CS equation fitted the experimental data very well, while the Taylor equation gave much smaller zero shear viscosity.

Because the zero shear viscosity of a polymer blend is a theoretical parameter, its determination by an experimental method may result in deviation from the theoretical

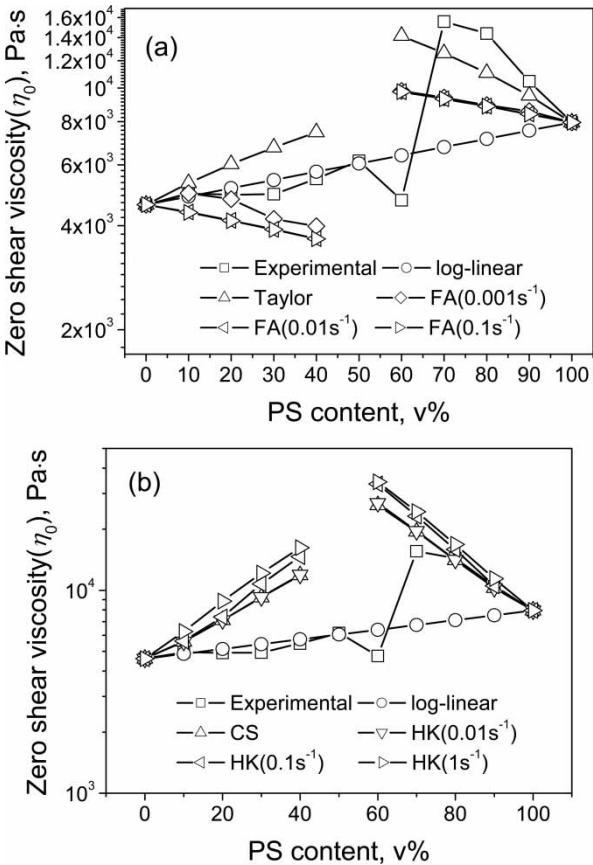


Figure 7. Fitted results from the Talyor, FA, CS, and HK equations.

value which can not be avoided completely. Different experimental methods using different equations to fit the experimental data would provide different deviations. The obvious differences between the theoretical equations and the experimental data in Fig. 7a,b may be due to the effect of the shear conditions applied to the blends. That is to say, the zero shear conditions used to determine the zero shear viscosities of the PP/PS blends have effects on the related phase structures, which makes it impossible to obtain the theoretical value of the zero shear viscosities.

When shear conditions were considered, the FA and HK equations were developed on the basis of the Taylor and CS equations, respectively. In Fig. 7a,b, the FA and HK equations were also applied to the PP/PS blends. Here it was very interesting that when the shear rate was increased for the FA equation, the fitted data decreased compared with the Taylor equation for all PP/PS blends with PS content of 0 to 100 vol. % (Fig. 7a), but the fitted data from the HK equation increased compared with the CS equation. Additionally and importantly, all four equations of Taylor, CS, FA and HK can not cover the whole experimental curve either when PS was dispersed in a PP phase, or when PP was dispersed in a PS phase.

3.4 Interpretations of Experimental Compositional Dependence of η_0

The above theoretical analysis showed that the compositional dependence of the zero shear viscosity of PP/PS blends was so complex that there were no suitable theoretical equations that can fit it well enough. But the above equations contained important information about polymer blends, which was helpful to understand the rheological behavior of our PP/PS blends.

Compared with blends in which PS was dispersed in a PP phase in the shape of spheres, blends with PP being dispersed in PS showed different rheological behaviors. But from the SEM images (Fig. 1) and the obtained d_p or S_p data (Fig. 2), the blends with PS or PP as dispersed phases had similar geometric phase structures. So there must be other important factors which impose a great effect on zero shear viscosity of blends. Here the factors should be not only a parameter of ratio of viscosities of components, but also include some parameters which are closely related to the response of the phase structures of the PP/PS blends to the shear conditions.

In Fig. 8, when PS is dispersed in PP, the deformation of hard PS spheres was smaller than that of the PP matrix. Thus a gap may be formed between the two phases at the interfaces and the hard PS spheres may rotate under shear conditions. On the other hand, for blends with soft PP spheres dispersed in a hard PS phase, the spheres were more deformable than the matrix. Thus gaps would not appear in Fig. 8c–d, which is obviously different from Fig. 8a,b. If the shear rate was large enough or the soft spheres were small enough, gaps may be formed mainly due to the extensive elongation of the hard PS matrix and the shape-keeping effect of the dispersed PP droplets due to the interfacial tension. Thus, compared with blends with PS dispersed in a PP matrix, PP/PS blends with PP dispersed in a PS phase provided less chance for interlayer slip between phases and were more sensitive to shear conditions.

In the Utracki equation, interlayer slip results in negative deviation and emulsion-like effects result in positive deviation. Because blends with PS dispersed in a PP matrix tended to generate interlayer slip easily by rotation of PS spheres, this was the predominant effect compared with the emulsion-like effect when shear rate sweeps were executed and then zero shear viscosities were calculated; these blends showed negative deviation. On the other hand, for blends with PP dispersed in a PS phase, interlayer slip can not be

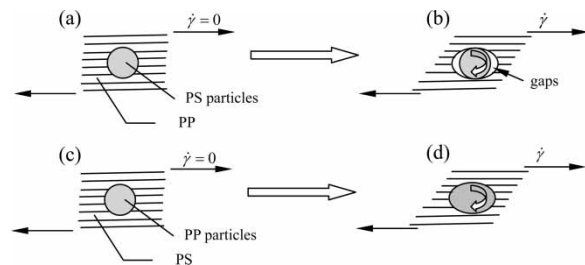


Figure 8. Schematic drawing of phase deformation in PP/PS blends with typical droplet-in-matrix phase structure to shear conditions.

engendered easily under low shear conditions, resulting in preponderance of the emulsion-like effect, under the effect of which the blends showed obvious positive deviation.

Compared with the compositional dependence of the calculated zero shear viscosity in Fig. 4, Figure 9 gives the compositional dependences at shear rates of 0.528 s^{-1} and 0.852 s^{-1} , showing that the viscosities of PP/PS blends decreased with increasing shear

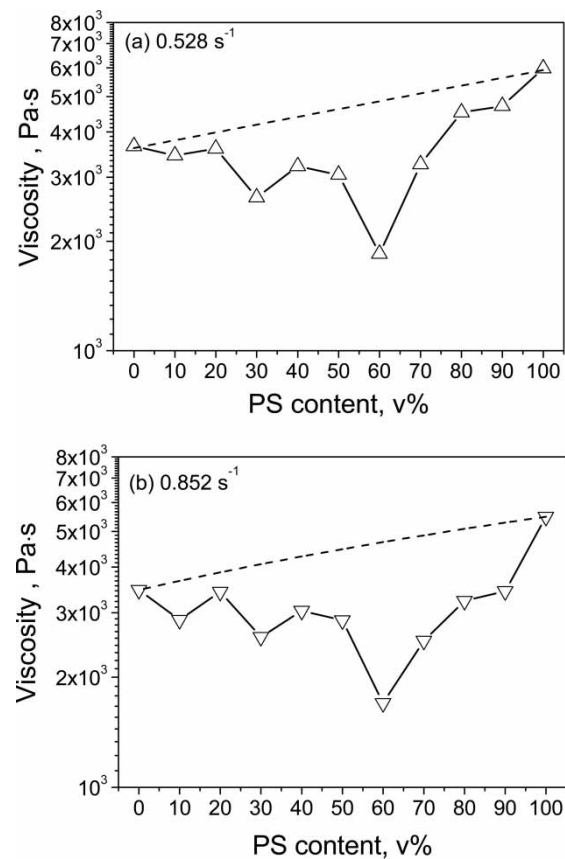


Figure 9. Dependence of PP/PS viscosity on composition at (a) 0.528 s^{-1} and (b) 0.852 s^{-1} .

rate. The effect of shear conditions on viscosities of blends was consistent with the Utracki and FA equations. Importantly, the viscosities of the blends were sensitive to the shear conditions when PP was dispersed in a PS continuous phase. This was consistent to the above interpretation on the basis of the sketch in Fig. 8.

In Fig. 9, it can also be seen that the sensitivity of blends to shear conditions increase with increasing content of dispersed phase. The blends with co-continuous phase structure showed the highest sensitivity, which implies that the interfacial slip was generated most easily. This viewpoint can be confirmed by the dependence of the critical shear rate ($\dot{\gamma}^*$) and power-law index (n) of PP/PS blends on composition, shown in Fig. 10a,b, where $\dot{\gamma}^*$ and n were both determined by fitting experimental data with the modified Carreau model and $\dot{\gamma}^* = \tau^*/\eta_0$. Smaller $\dot{\gamma}^*$ means the blend was sensitive to weaker shear conditions and was easier to fall into the non-Newtonian region. Smaller n represents stronger non-Newtonian characteristics and a strong dependence on shear conditions.

Compared with the compositional dependence of d_p or S_p of PP/PS blends (in Fig. 2), the variations with PS content of zero shear viscosities (in Fig. 4), viscosity at different shear rates (in Fig. 9), critical shear rate and power-law index (in Fig. 10) were more similar to d_p than to S_p . But the polymer blend system was so complex that it was difficult to provide a suitable theoretical equation for the compositional dependence of

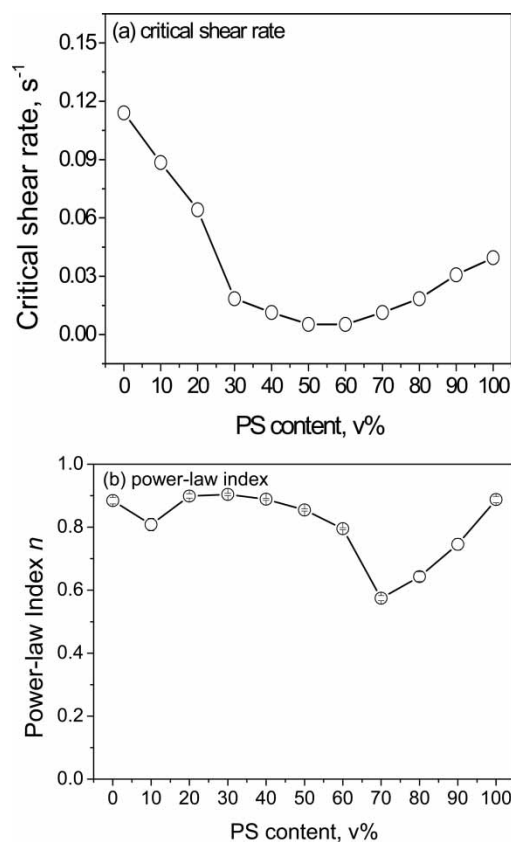


Figure 10. Dependence of power-law index n of PP/PS blends on PS content.

the zero shear viscosities. The mechanism of response of complex phase structures of polymer blends to shear conditions must be the key to understand better or find out the real relations between complex rheological behaviors and structures of blends.

4 Conclusion

After the phase structures of immiscible polypropylene(PP)/polystyrene(PS) blends with different compositions were observed by SEM, the average numerical diameter (d_p) and the specific interfacial area (S_p) were calculated according to the obtained SEM images. Then the rheological behaviors of PP/PS blends with different compositions were studied by static shear rate sweep at 190°C with a stress tech fluids rheometer. The zero shear viscosities of the blends were determined according to modified Carreau model. The results showed that the compositional dependence of zero shear viscosity of PP/PS deviated greatly from linear or log-linear additivity. When PS was dispersed in a PP continuous phase, the blends showed negative deviation, while for blends with PP dispersed in a PS matrix, larger positive deviation was generated. The different theoretical equations of Nielsen, Utracki, Taylor, FA, CS, and HK were used to fit the experimental data of the zero shear viscosities of the blends. It was found that none of them were suitable for PP/PS blends. To interpret the compositional dependence of zero shear viscosities of PP/PS blends, the hard PS spheres dispersed in PP matrix were presumed to rotate easily with gaps being formed between the two phases at the interfaces, resulting in more interlayer slip and then a larger negative deviation. On the other hand, for blends with soft PP dispersed in a hard PS matrix, it was difficult to generate interlayer slip under low shear conditions. Only under stronger shear conditions, or if the dispersed PP spheres were small enough, interlayer slips may occur due to the elongation of the PS matrix and shape-retention of PP spheres because of interfacial tension. The experimental data showed that blends with PP dispersed in PS were more sensitive to shear conditions compared with blends having similar geometric phase structures for the case that PS was dispersed in a PP matrix. In addition, with increasing content of dispersed phase, blends had higher sensitivity to shear conditions; no matter whether PP or PS was dispersed in the other polymer component.

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